

Lemma. Let  $F$  be a field,  $f(x) \in F[x]$  be a polynomial and  $D_x f(x)$  be the derivative.  
 $\alpha$  is a multiple root of  $f$  if and only if  $\alpha$  is a root of both  $f$  and  $D_x f$ .

In particular,

$f(x)$  is separable if and only if  $f(x)$  and  $D_x f(x)$  are relatively prime.

Proof. Let  $K$  be the splitting of  $f(x)$  and  $D_x f(x)$ .

Suppose that  $f$  has  $\alpha$  as a multiple root. That is,  $f(x)$  is divided by  $(x-\alpha)^2$ , so

$$f(x) = (x-\alpha)^2 \cdot g(x) \text{ for some } g \in K[x].$$

Taking the formal derivative to  $f$ , then

$$D_x f(x) = 2 \cdot (x-\alpha) \cdot g(x) + (x-\alpha) \cdot D_x g(x)$$

clearly it has  $\alpha$  as a root. Conversely,  $\alpha \in K$  is root of  $f$  and  $D_x f$ .

Then, we can represent  $f(x) = (x-\alpha) \cdot h(x)$ , and taking derivative then

$$D_x f(x) = h(x) + (x-\alpha) \cdot D_x h(x).$$

Since  $D_x f(\alpha) = 0$  by the assumption, so  $h(\alpha) = 0$ . It is proved the first assertion.

Now, the second assertion is shown directly:

$f(x)$  is separable iff  $f(x)$  has no multiple root iff  $f(x)$  and  $D_x f(x)$  are relatively prime.

Note: To clarify understand, consider the negation.  $\square$

Corollary. Every irreducible polynomial over a field of characteristic 0 is separable.

Proof. Let  $F$  be a field of characteristic 0, and  $f(x) \in F[x]$  be an irreducible.

Then,  $D_x f(x)$  is polynomial of degree  $\deg f - 1$ , so therefore it must be relatively prime.  $\square$

Note: But, in the field of characteristic  $p \neq 0$ ,  $D_x f(x)$  can be zero,

Thm. A field of characteristic 0 is perfect.

Proof. Let  $f(x) \in F[x]$  be an irreducible polynomial. If  $D_x f(x)$  is not zero, then  $\deg D_x f < \deg f$ .

Therefore,  $(f, D_x f) = 1$ , because if  $(f, D_x f) \neq 1$ , then it is contradiction to  $f$  is irreducible. So it is separable.

Lemma. A field of characteristic  $p$  is perfect iff every element of  $F$  has a  $p$ th root in  $F$ .

Proof. Let  $F$  be a field of characteristic  $p$ .

Suppose that  $F$  is perfect. Let  $\alpha \in F$  be a non-zero, non-1. put  $f(x) = x^p - \alpha$ .

If  $f(x)$  has a root in  $F$ , then there is nothing to prove.

Assume  $f$  has no root in  $F$ . Observe that in splitting field  $K$  of  $f(x)$ , put  $\beta \in K$  s.t.

$$f(\beta) = \beta^p - \alpha = 0.$$

And,  $f(x) = x^p - \alpha = x^p - \beta^p = (x - \beta)^p$  in  $K$ . But, if  $f$  is reducible in  $F$ , then for some  $1 \leq k < p$ ,  $f(x) = (x - \beta)^{p-k} \cdot (x - \beta)^k$  and  $(x - \beta)^{p-k}, (x - \beta)^k \in F[x]$ .

That is,  $(-\beta)^k \in F$ . Clearly,  $(k, p) = 1$ , thus  $ku + pv = 1$  for some  $u, v \in \mathbb{Z}$ .

Now,  $\beta = \beta^{ku+pv} = (\beta^k)^u \cdot (\beta^p)^v = (\beta^k)^u \cdot \alpha^v \in F$ . It is contradiction. But, if  $f$  is irreducible then  $D_x f = p \cdot x^{p-1} = 0$  implies contradiction with  $F$  is perfect. Now,  $\alpha$  has a  $p$ th root in  $F$ .

Conversely, let  $f(x) \in F[x]$  be an irreducible. Assume it is inseparable.

Denote  $f(x) = a_n x^n + \dots + a_1 x + a_0$ . Since  $f(x)$  is irreducible and inseparable,

$D_x f(x) = n a_n x^{n-1} + (n-1) a_{n-1} x^{n-2} + \dots + a_1$  must be zero, that is,

$$a_k \neq 0 \Leftrightarrow k \in (p)$$

Therefore, we can write

$$f(x) = b_m x^{mp} + b_{m-1} x^{(m-1)p} + \dots + b_1 x^p + b_0$$

Now, by the assumption, each  $b_i$  has  $p$ th root  $c_i$ , thus

$$f(x) = c_m^p (x^m)^p + c_{m-1}^p (x^{m-1})^p + \dots + c_0^p$$

$$= (c_m x^m + c_{m-1} x^{m-1} + \dots + c_0)^p. \text{ But it is reducible. Contradiction.}$$

Corollary. Every finite field is perfect. (char  $p$  is 1 on finite, thus Auto.)

Thm. Every finite separable extension is simple extension.

Proof. If  $F$  is finite, then the finite extension  $E/F$  is also finite. Then,  $E^X$  is cyclic, therefore  $E = F(\alpha)$  where  $\alpha$  is generator of  $E^X$ .

Now, assume that  $F$  is infinite. Since  $E/F$  is finite,

$$E = F(\alpha_1, \dots, \alpha_r) \text{ where } \alpha_1, \dots, \alpha_r \in E.$$

Now, enough to show  $F(\beta, \gamma) = F(\alpha)$  for some  $\alpha \in E$ , and after is solved by the induction.

Let  $\beta_1, \dots, \beta_n$  be distinct zeros of  $\text{irr}(\beta, F)$ , and

$\gamma_1, \dots, \gamma_m$  be distinct zeros of  $\text{irr}(\gamma, F)$ , in  $\bar{F}$ . ( $\beta_1 = \beta$ ,  $\gamma_1 = \gamma$ ).

Since  $E/F$  is separable, each zeros are multiplicity 1.

Take  $\alpha \in F$  s.t.  $\alpha \neq \frac{\beta_i - \beta}{\gamma - \gamma_j}$  for all  $1 \leq i \leq n$ ,  $2 \leq j \leq m$ , since  $F$  is infinite.

Put  $\alpha = \beta + \alpha \cdot \gamma$ . Then, for all  $(i, j) \in \{1, \dots, n\} \times \{2, \dots, m\}$ ,

$$\alpha = \beta + \alpha \gamma \neq \beta_i + \alpha \gamma_j, \text{ or } \alpha - \alpha \gamma_j \neq \beta_i.$$

Observe that:  $\text{irr}(\beta, F)(\alpha - \alpha \gamma) \in F(\alpha)[x]$ , and

$$\text{irr}(\beta, F)(\alpha - \alpha \gamma) = \text{irr}(\beta, F)(\beta) = 0.$$

But,  $\text{irr}(\beta, F)(\alpha - \alpha \gamma_j) \neq 0$  for all  $2 \leq j \leq m$ ,

That is,  $\text{irr}(\beta, F)(\alpha - \alpha \gamma)$  and  $\text{irr}(\beta, F)(x)$  have a common factor  $(x - \gamma)$  in  $F(\alpha)$ , thus  $\gamma \in F(\alpha)$ , so  $\beta = \alpha - \alpha \gamma \in F(\alpha)$ .

